Exact Wiener–Hopf Solution of the Toda Fermi-Edge Limit, Two-Loop $O(1/Y)$ Asymptotics, and KP Master-Function Synthesis

Project *quantum_boussinesq_solitons* (Phase 2i)

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Abstract

We carry out Phase 2i of the programme while incorporating and resolving the mathematical findings of the six independent Phase 2h and follow-up audits (`independent_phase2h_audit.pdf`, `audit_charges_action.pdf`, `audit_determinant_foundations.pdf`, `audit_frames_defects.pdf`, `audit_toda_dictionary.pdf`, and `audit_projected_measure.pdf`):

1. **Exact Wiener–Hopf solution of the Toda Fermi-edge Bethe integral equation (c):** In the Fermi-edge double-scaling limit $p = q + \hbar_T u$, the Toda Bethe operator on the semi-infinite Fermi sea $(-\infty, 0)$ has canonical Wiener–Hopf factor $\widetilde{\Phi}_{\mathrm{WH}}(s) = s^{-1/2}\Gamma(1+\frac{s}{2\pi})\exp(-\frac{s}{2\pi}\ln\frac{s}{2\pi e})$. The Laplace transform of the dressed momentum $\frac{4}{3}y(u)^{3/2}$ equals $\widetilde{\Phi}_{\mathrm{WH}}(s)/s^2$, yielding the exact asymptotic expansion of the quasi-momentum shift $\delta u(y) = u(y) - y$:

$$\delta u(y) = -\frac{\ln y}{2\pi} + c_{\mathrm{edge}} + \frac{d_0}{y} + \frac{e_0}{y^2} + O(y^{-3}),$$

   where all $(\ln y)/y$, $(\ln^2 y)/y^2$, and $(\ln y)/y^2$ coefficients vanish identically ($d_1 = e_2 = e_1 = 0$), and $c_{\mathrm{edge}} = \frac{1-\ln(8\pi)}{2\pi}$, $d_0 = \frac{1}{48} - \frac{1}{4\pi^2}$, $e_0 = \frac{4+6\zeta(3)-\pi^2}{96\pi^3}$.

2. **Analytical proof of $R_0(\infty, r) = -\frac{4}{\pi}(\chi_2(r) - r)$, cancellation of $\frac{\ln^2 Y}{Y}$ and $\frac{\ln Y}{Y}$, and closed-form two-loop coefficient $c_1(r)$ (c):** For two excitations $u_1 > u_2 > 0$ outside the Fermi sea, the double Laplace transform of $\partial_{u_2}\Phi_q(u_1, u_2)$ is $2\frac{\widetilde{\Phi}_{\mathrm{WH}}(s_1)\widetilde{\Phi}_{\mathrm{WH}}(s_2)-1}{s_1+s_2}$, giving the Krein–Wiener–Hopf convolution $\Phi_q(u(Y), u(r^2Y)) = 4\int_0^{r^2Y}\psi_-(u(Y) - u(r^2Y) + u(\eta))\sqrt{\eta}\,d\eta$. Because the argument shift inside ψ_- is the balanced 4-term coboundary $\Delta(Y,t) = \delta u(Y) - \delta u(r^2Y) + \delta u(Yt^2) - \delta u(Y(1-r^2+t^2))$ and $d_1 = 0$, all $\ln Y$ and c_{edge} terms cancel identically at both $O(Y^0)$ and $O(Y^{-1})$. The leading limit is proved analytically from the integral equation to be $R_0(\infty, r) = -\frac{4}{\pi}(\chi_2(r) - r)$, and the $O(1/Y)$ (two-loop) coefficient evaluates in closed elementary form to

$$c_1(r) = -\frac{r^3}{6(1-r^2)} - 2d_0\left(\frac{1-r^2}{r^2}\ln\frac{1-r}{1+r} + \frac{2}{r}\right).$$

   Subtracting the exact finite-h classical Toda lattice dispersion phase $\delta_T(\kappa_1, \kappa_2)/h$ from the saved Phase 2f data removes the $O(Y^2h + Y^3h^2)$ curvature that distorted the $h\ln h$ edge fit in `matching_analysis.json`, matching the exact $c_1(r)$ to $< 2 \times 10^{-4}$.

3. **Inversion-antisymmetric KP master kernel $\widetilde{K}_L(z)$, bootstrap synthesis, and audit resolutions (c)/(d):** We prove that $\mathcal{F}_K(p,q) = -\mathcal{F}_K(q,p)$ and $\mathcal{I}_{SS}[K](1,2) = -\mathcal{I}_{SS}[K](2,1)$ hold iff $K(-z) = -K(z)$ and $K(1/z) = -K(z)$. Replacing $K_L(z)$ by $\widetilde{K}_L(z) = \frac{K_L(z) - K_L(1/z)}{2} = K_L(z) + \frac{\pi}{8}\,\mathrm{sgn}(\mathrm{Re}\,z)$ resolves the $\epsilon_1 = -1$ orientation discrepancies, gives exact crossing symmetry and Darboux-shifted defect fusion, and continues holomorphically to the breather locus with $\mathrm{Re}\,\mathcal{I}_{SS}[\widetilde{K}_L](\bar{\alpha} - i\gamma, \bar{\alpha} + i\gamma) = 0$. Remaining open field-theoretic questions remain explicitly labelled **(d)**.

All **(c)** claims are verified in `scripts/toda_wh_and_kp_synthesis.py` (`ALL CHECKS PASSED`).

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1 Exact Wiener–Hopf Solution of the Toda Fermi-Edge Limit

1.1 Wiener–Hopf Factorisation on the Semi-Infinite Fermi Sea

In `scripts/toda_edge_ba.py`, the ground-state density $\rho(p)$ and phase-shift dressing function $F(p|p_2)$ on $[-q,q]$ satisfy

$$2\pi h\,\rho(p)-\int_{-q}^{q}K(p-p')\,\rho(p')\,dp'=1,\qquad 2\pi h\,F(p|p_2)-\int_{-q}^{q}K(p-p')\,F(p'|p_2)\,dp'=-\Theta(p-p_2), \tag{1}$$

where $h=\hbar_T=3b\hbar_B$ ($a=1$, $b=e^{-1/2}$) and

$$\Theta(D)=2h\,\mathrm{Im}\ln\Gamma\!\left(1+\frac{iD}{h}\right)+2D\ln h,\qquad K(D)=\Theta'(D)=2\,\mathrm{Re}\,\psi\!\left(1+\frac{iD}{h}\right)+2\ln h. \tag{2}$$

In the Fermi-edge scaling limit $p=q+hx$ ($h\to 0$ at fixed x), the left edge $-q$ recedes to $x=-2q/h\to-\infty$, so the Fermi sea occupies the negative half-line $x'\in(-\infty,0)$ and the integral operator becomes a Wiener–Hopf operator with kernel $K_0(x-x')=2\,\mathrm{Re}\,\psi(1+i(x-x'))$.

Theorem 1.1 (Wiener–Hopf factorisation and dressed momentum, **(c)**, `toda_wh_and_kp_synthesis.py`). *1*

The Fourier symbol of $2\pi\delta(x)-K_0(x)$ is

$$\Sigma_q(\omega):=\int_{-\infty}^{\infty}\big[2\pi\delta(x)-2\,\mathrm{Re}\,\psi(1+ix)\big]e^{-i\omega x}\,dx=\frac{2\pi^2}{\sinh(\pi|\omega|)}e^{\pi|\omega|}=\Phi_{\mathrm{WH}}(i\omega)\,\Phi_{\mathrm{WH}}(-i\omega), \tag{3}$$

where the canonical Wiener–Hopf factor analytic and non-vanishing in $\mathrm{Re}\,s>0$ ($s=-i\omega$) is

$$\Phi_{\mathrm{WH}}(s)=\sqrt{\frac{2\pi}{s}}\,\Gamma\!\left(1+\frac{s}{2\pi}\right)\exp\!\left(-\frac{s}{2\pi}\ln\frac{s}{2\pi e}\right),\qquad \widetilde{\Phi}_{\mathrm{WH}}(s):=\frac{\Phi_{\mathrm{WH}}(s)}{\sqrt{2\pi}}\underset{s\to+\infty}{\longrightarrow}1. \tag{4}$$

2. *Matching the homogeneous Wiener–Hopf edge density to the bulk classical arcsine asymptote $\rho_{\mathrm{cl}}(q-hz)\sim\frac{1}{2\pi a\sqrt{bhz}}$ ($z\to+\infty$) yields the exact Laplace transform of the scaled dressed momentum $\lim_{h\to0}\frac{a\,[P_{\mathrm{dr}}(q+hu)-P_{\mathrm{dr}}(q)]}{\sqrt{bh}\,h}=\frac{4}{3}y(u)^{3/2}$ for $u>0$:*

$$\int_0^{\infty}\left(\frac{4}{3}y(u)^{3/2}\right)e^{-su}\,du=\frac{\widetilde{\Phi}_{\mathrm{WH}}(s)}{s^2}=s^{-5/2}\,\Gamma\!\left(1+\frac{s}{2\pi}\right)\exp\!\left(-\frac{s}{2\pi}\ln\frac{s}{2\pi e}\right). \tag{5}$$

3. *Expanding (5) around the branch point $s=0$ via $\ln\Gamma(1+\frac{s}{2\pi})=-\gamma_E\frac{s}{2\pi}+\frac{\pi^2}{12}(\frac{s}{2\pi})^2-\frac{\zeta(3)}{3}(\frac{s}{2\pi})^3+\cdots$ and inverting term-by-term via $\mathcal{L}^{-1}[s^{-\alpha}(\ln s)^k](u)=(-1)^k\partial_\alpha^k[u^{\alpha-1}/\Gamma(\alpha)]$ gives as $u\to+\infty$:*

$$y(u)-u=\frac{\ln(8\pi u)-1}{2\pi}+\frac{12\ln(8\pi u)-\pi^2}{48\pi^2\,u}+\frac{-12\ln(8\pi u)+6\zeta(3)+\pi^2-8}{96\pi^3\,u^2}+O(u^{-3}). \tag{6}$$

4. *Inverting (6) for $\delta u(y) := u(y) - y$ as $y \to +\infty$, all $\frac{\ln y}{y}$, $\frac{\ln^2 y}{y^2}$, and $\frac{\ln y}{y^2}$ terms cancel identically ($d_1 = e_2 = e_1 = 0$), leaving the pure power-tail expansion after the leading logarithm:*

$$\boxed{\delta u(y) = -\frac{\ln y}{2\pi} + c_{\text{edge}} + \frac{d_0}{y} + \frac{e_0}{y^2} + O(y^{-3}),} \tag{7}$$

with exact analytical coefficients

$$c_{\text{edge}} = \frac{1 - \ln(8\pi)}{2\pi} = -0.35398787697503518285\ldots, \tag{8}$$

$$d_0 = \frac{1}{48} - \frac{1}{4\pi^2} = -0.00449696257725110953\ldots, \tag{9}$$

$$e_0 = \frac{4 + 6\zeta(3) - \pi^2}{96\pi^3} = +0.00045109717881779249\ldots \tag{10}$$

Remark 1.2 (**(c)**). In (9), $\frac{1}{48} = \frac{B_2}{8}$ originates from the two-loop Stirling coefficient $B_2 = \frac{1}{6}$ of $\ln\Gamma(1 + \frac{s}{2\pi})$, while $-\frac{1}{4\pi^2}$ originates from the square of the one-loop Wiener–Hopf dressing shift. Table 1 compares the analytical expansion (7) against polynomial $h \to 0$ extrapolations of `FastNodesEdgeBA` after subtracting the exact classical root shift $2b(\cosh\kappa - 1)$.

Table 1: Verification of the exact Wiener–Hopf expansion (7) of $\delta u(y)$ against cubic $h \to 0$ extrapolations ($h \in \{0.04, 0.02, 0.01, 0.005\}$) of `FastNodesEdgeBA` **(c)**.

y	$\delta u_{\text{num}}(y)$	$\delta u_{\text{WH}}(y)$ (to $O(y^{-2})$)	|diff|	$y\big(\delta u_{\text{num}}(y) + \frac{\ln y}{2\pi} - c_{\text{edge}}\big)$	$d_0 + e_0/y$
2.0	−0.46639236	−0.46644138	4.9×10^{-5}	−0.004173	−0.004271
4.0	−0.57571472	−0.57571952	4.8×10^{-6}	−0.004365	−0.004384
8.0	−0.68549578	−0.68549635	5.7×10^{-7}	−0.004436	−0.004441
16.0	−0.79553827	−0.79553838	1.1×10^{-7}	−0.004467	−0.004469
32.0	−0.90571661	−0.90571697	3.5×10^{-7}	−0.004472	−0.004483

1.2 Krein–Wiener–Hopf Convolution for the Two-Particle Dressed Phase

We now derive the two-particle Fermi-edge dressed phase

$$\Phi_q(u_1, u_2) := \lim_{h \to 0} \frac{\Phi(q + hu_1 \,|\, q + hu_2) - \Phi(q + hu_1 \,|\, q)}{h} \qquad (u_1 > u_2 > 0) \tag{11}$$

directly from the Bethe integral equation, resolving Section 3.2 and Section 4 of `independent_phase2h_audit.pdf` and Section 4 of `audit_frames_defects.pdf`.

Theorem 1.3 (Krein convolution and large-Y expansion of $R_0(Y, r)$, **(c)**, `toda_wh_and_kp_synthesis.py`)**.** *For $u_1, u_2 > 0$ outside the Fermi sea, the Wiener–Hopf projection onto $x > 0$ yields the exact double Laplace transform*

$$\int_0^\infty du_1\, e^{-s_1 u_1} \int_0^\infty du_2\, e^{-s_2 u_2}\, \frac{\partial \Phi_q(u_1, u_2)}{\partial u_2} = 2\, \frac{\widetilde{\Phi}_{\text{WH}}(s_1)\, \widetilde{\Phi}_{\text{WH}}(s_2) - 1}{s_1 + s_2}. \tag{12}$$

For $u_1 > u_2 > 0$ (where $\mathcal{L}^{-1}[(s_1 + s_2)^{-1}] = \delta(u_1 - u_2) = 0$), integrating with respect to u_2 from the edge $u_2 = 0$ ($\Phi_q(u_1, 0) = 0$) gives the exact convolution

$$\boxed{\Phi_q(u_1, u_2) = 2 \int_0^{u_2} \psi_-(u_1 - w)\, \widetilde{\Psi}_-(u_2 - w)\, dw = 4 \int_0^{r^2 Y} \psi_-\big(u(Y) - u(r^2 Y) + u(\eta)\big)\, \sqrt{\eta}\, d\eta,} \tag{13}$$

where $\widetilde{\Psi}_-(u) = \frac{d}{du}\big(\frac{4}{3} y(u)^{3/2}\big) = 2\sqrt{y(u)}\, y'(u)$, $\psi_-(u) = \widetilde{\Psi}'_-(u)$, and in the second equality we changed integration variable to the dressed momentum $v = u_2 - w = u(\eta)$ with $\eta \in [0, r^2 Y]$.

2. *Setting $\eta = Yt^2$ ($t \in [0,r]$) and $\eta_0(t) := Y(1-r^2+t^2) =: Y\sigma(t)$, the argument of ψ_- in (13) is $u(\eta_0(t)) + \Delta(Y,t)$, where $\Delta(Y,t)$ is the balanced 4-term coboundary*

$$\Delta(Y,t) = \delta u(Y) - \delta u(r^2Y) + \delta u(Yt^2) - \delta u\big(Y\sigma(t)\big) = \underbrace{-\frac{1}{2\pi}\ln\frac{t^2}{r^2\sigma(t)}}_{\Delta_0(t)} + \frac{d_0}{Y}\underbrace{\left(1 - \frac{1}{r^2} + \frac{1}{t^2} - \frac{1}{\sigma(t)}\right)}_{\Delta_1(t)} + O(Y^{-2}), \tag{14}$$

*in which both $-\frac{\ln Y}{2\pi}$ and c_{edge} cancel identically. Moreover, because $d_1 = 0$, $\psi_-(u(\eta)) = \big(\frac{1}{u'(\eta)}\frac{d}{d\eta}\big)^2(\frac{4}{3}\eta^{3/2}) = \eta^{-1/2}\big[1-(2d_0+\frac{3}{4\pi^2})\eta^{-2}+O(\eta^{-3})\big]$ contains neither $\ln\eta$ nor c_{edge}. Consequently, $\Phi_q(u(Y),u(r^2Y))$ has **no** $\ln Y$ or c_{edge} terms at $O(Y^0)$ or $O(Y^{-1})$: the $\frac{1-r^2}{4\pi^2 r}\frac{\ln^2 Y}{Y}$ and $\frac{\ln Y}{Y}$ terms in the bare classical Hessian $Q = \frac{(\delta u_2 - r^2\delta u_1)^2}{r(1-r^2)Y}$ are cancelled identically by opposite terms in $d\Phi_{\text{loop}}(Y,r) = \Phi_q(u_1,u_2) - \Phi_{\text{cl}}(u_1,u_2)$.*

3. *Under the rationalising substitution $s = \frac{t}{\sqrt{1-r^2+t^2}} \in [0,r]$ ($\Delta_0(t) = -\frac{1}{\pi}\ln\frac{s}{r}$, $\frac{t^2\,dt}{\sigma(t)^{3/2}} = (\frac{1}{1-s^2}-1)\,ds$), the $O(Y^0)$ integral gives in one line the analytical proof of the leading residual:*

$$\boxed{\lim_{Y\to\infty} R_0(Y,r) = -4\int_0^r \Delta_0(t)\frac{t^2\,dt}{(1-r^2+t^2)^{3/2}} = \frac{4}{\pi}\int_0^r \ln\!\Big(\frac{s}{r}\Big)\left(\frac{1}{1-s^2}-1\right)ds = -\frac{4}{\pi}\big(\chi_2(r)-r\big).} \tag{15}$$

4. *At $O(1/Y)$ (two loops in the Boussinesq/KdV scaling limit), the Wiener–Hopf integral evaluates in closed elementary form to*

$$\boxed{R_0(Y,r) = -\frac{4}{\pi}\big(\chi_2(r)-r\big) + \frac{c_1(r)}{Y} + O(Y^{-2}), \qquad c_1(r) = -\frac{r^3}{6(1-r^2)} - 2d_0\left(\frac{1-r^2}{r^2}\ln\frac{1-r}{1+r} + \frac{2}{r}\right),} \tag{16}$$

where $d_0 = \frac{1}{48} - \frac{1}{4\pi^2}$. Relative to the bare two-body Gamma-ratio Stirling coefficient $c_1^{\Gamma}(r) = -\frac{r^2}{6(1-r^2)}$ of `smatrix_ansatz.tex`*, the $\frac{1}{1-r}$ threshold pole cancels identically in the collective Fermi-sea dressing shift:*

$$\Delta c_1(r) := c_1(r) - c_1^{\Gamma}(r) = \frac{r^2}{6(1+r)} - 2d_0\left(\frac{1-r^2}{r^2}\ln\frac{1-r}{1+r} + \frac{2}{r}\right), \qquad \lim_{r\to 1^-}\Delta c_1(r) = \frac{1}{\pi^2}. \tag{17}$$

Remark 1.4 (Resolution of `audit_frames_defects.pdf` Section 4, `audit_toda_dictionary.pdf` Section 3, and
As correctly pointed out in Section 4 of `audit_frames_defects.pdf`, the subtracted edge term $-\Phi(p_1|e_2)/h$ in $\Psi_{2,T}$ depends only on $y_1 = Y$ and not on $y_2 = r^2Y$, so $\partial_r\Phi(p_1|e_2) = 0$; its sole role in (13) is to fix the lower integration limit $u_2 = 0$ ($\Phi_q(u_1,0) = 0$). Equation (15) shows where the $+\frac{4}{\pi}r$ term in $R_0(\infty,r)$ actually comes from: it is the integral $-\frac{4}{\pi}\int_0^r \ln(s/r)\,ds = +\frac{4}{\pi}r$ of the -1 part of $\frac{s^2}{1-s^2} = \frac{1}{1-s^2} - 1$ inside the u_2-dependent dressed phase convolution, i.e. the one-loop counterpart of the $+4rY$ classical frame cancellation.

2. Directly connecting to the exact endpoint reciprocity identity of `audit_toda_dictionary.pdf` Eqs. (6)–(12), differentiating (13) gives $(\partial_{u_1} + \partial_{u_2})\Phi_q(u_1,u_2) = 2\psi_-(u_1)\widetilde{\Psi}_-(u_2)$, so the four-term symmetric sum in the Fermi-edge limit $h \to 0$ is the factorised endpoint product

$$\widehat{\Phi}_q\big(u(y_1),u(y_2)\big) + \widehat{\Phi}_q\big(u(y_2),u(y_1)\big) = 2\widetilde{\Psi}_-\big(u(y_1)\big)\widetilde{\Psi}_-\big(u(y_2)\big) = \frac{8\sqrt{y_1y_2}}{u'(y_1)\,u'(y_2)}.$$

Subtracting the classical frame sum $-8\sqrt{y_1y_2}$ yields the exact edge limit of the symmetric remainder S_{rem}/h of `audit_toda_dictionary.pdf` Section 4:

$$\lim_{h\to 0}\frac{S_{\text{rem}}(Y,r^2Y)}{h} = 8rY\left(\frac{1}{u'(Y)\,u'(r^2Y)} - 1\right) = \frac{4}{\pi}\left(r + \frac{1}{r}\right) + \frac{1}{Y}\left[\frac{2+8\pi^2 d_0}{\pi^2}\left(r + \frac{1}{r^3}\right) + \frac{2}{\pi^2 r}\right] + O(Y^{-2}).$$

The $O(Y^0)$ term $\frac{4}{\pi}(r+\frac{1}{r})$ is precisely the symmetric sum of the $+\frac{4}{\pi}r$ term in $R_0(\infty,r)$ and its $(1\leftrightarrow 2)$ partner $+\frac{4}{\pi}(1/r)$, arising from the quantum Jacobian $u'(y)=1-\frac{1}{2\pi y}-\frac{d_0}{y^2}+O(y^{-3})\neq 1$.

3. In `matching_toda.py`, the raw residual subtracted $2Y(1-r^2)\ln\frac{1-r}{1+r}$ rather than the finite-h classical Toda phase $\delta_T(\kappa_1,\kappa_2)/h$, leaving a classical lattice dispersion series $a_1^{\rm cl}(r)Y^2h+a_2^{\rm cl}(r)Y^3h^2+O(Y^4h^3)$. Because $p_j-q(h)$ is measured from the quantum Fermi edge $q(h)$ (which already absorbs the $-\frac{1}{2\pi}h\ln h$ shift), the finite-h error at fixed (Y,r) is a smooth power series in h. When `matching_analysis.py` fitted $1,h,h\ln h$ to three points, the spurious $h\ln h$ basis function absorbed the $O(Y^3h^2)$ classical curvature and flipped the sign of $\widehat{c}_1(r)$. Subtracting $\delta_T(\kappa_1,\kappa_2)/h$ from the saved `matching_toda_data.json` and `matching_toda_fine.json` phases yields $c_1^{\rm num}(0.4,0.5,0.6)=(-0.00783,-0.02157,-0.04861)$, agreeing with the exact formula (16) $(-0.00774,-0.02144,-0.04844)$ to $<2\times10^{-4}$ (Table 2).

Table 2: Exact one-loop limit $R_0(\infty,r)$, two-loop coefficient $c_1(r)=c_1^{\rm rat}(r)+c_1^{(1/\pi^2)}(r)$, bare Gamma-ratio coefficient $c_1^{\Gamma}(r)$, collective dressing difference $\Delta c_1(r)$, and comparison with classical-subtracted saved Phase 2f Toda data **(c)**.

r	$R_0(\infty,r)$	Exact $c_1(r)$	$c_1^{\rm rat}(r)$	$c_1^{(1/\pi^2)}(r)$	$\Delta c_1(r)$	Saved Toda $c_1^{\rm num}(r)$
0.20	−0.00114841	+0.00102901	−0.01259045	+0.01361946	+0.00797346	—
0.30	−0.00394949	−0.00128010	−0.02192391	+0.02064381	+0.01520342	—
0.40	−0.00962286	−0.00773659	−0.03568534	+0.02794875	+0.02400944	−0.007827
0.50	−0.01951541	−0.02144459	−0.05711791	+0.03567332	+0.03411097	−0.021566
0.60	−0.03545557	−0.04843599	−0.09245042	+0.04401443	+0.04531401	−0.048613
0.70	−0.06018350	−0.10263222	−0.15591408	+0.05328186	+0.05749850	—
0.80	−0.09840626	−0.22566817	−0.28970625	+0.06403809	+0.07062813	—

2 KP Master Function Synthesis, Bootstrap Constraints, and Audit Resolutions

2.1 Exchange-Antisymmetry Criterion and the Subtracted Kernel $\widetilde{K}_L(z)$

We now resolve Section 2 of `independent_phase2h_audit.pdf` and Section 2 of `audit_determinant_foundation`

Theorem 2.1 (Inversion-antisymmetry and 4-orientation identities, **(c)**, `toda_wh_and_kp_synthesis.py`)**.**

*Under particle exchange $(1\leftrightarrow 2)$ or KP-vertex swap $(p\leftrightarrow q)$, every root-ratio argument $z=w_r(\alpha_1,\epsilon_1)/w_s(\alpha_2,\epsilon_2)$ inverts: $z\mapsto 1/z$. Consequently, $\mathcal{F}_K(p,q)=-\mathcal{F}_K(q,p)$ and $\mathcal{I}_{SS}[K](1,2)=-\mathcal{I}_{SS}[K](2,1)$ hold universally if and only if $K(z)$ is **both odd** $(K(-z)=-K(z))$ **and inversion-antisymmetric** $(K(1/z)=-K(z))$.*

2. *For the one-loop kernel $K_L(z)=-\frac{1}{\pi}\chi_2(z)-\frac{1}{2\pi}\ln|z|\ln\left|\frac{1-z}{1+z}\right|$, Euler's identity $\chi_2(z)+\chi_2(1/z)=\frac{\pi^2}{4}\operatorname{sgn}(\operatorname{Re}z)$ yields the inversion anomaly $K_L(z)+K_L(1/z)=-\frac{\pi}{4}\operatorname{sgn}(\operatorname{Re}z)$. Defining the inversion-antisymmetrised kernel*

$$\boxed{\widetilde{K}_L(z):=\frac{K_L(z)-K_L(1/z)}{2}=K_L(z)+\frac{\pi}{8}\operatorname{sgn}(\operatorname{Re}z)} \tag{18}$$

makes $\widetilde{K}_L(-z)=-\widetilde{K}_L(z)$ and $\widetilde{K}_L(1/z)=-\widetilde{K}_L(z)$ exact on all of $\mathbb{C}\backslash i\mathbb{R}$, so that $\mathcal{F}_{\widetilde{K}_L}(p,q)=-\mathcal{F}_{\widetilde{K}_L}(q,p)$ and $\mathcal{I}_{SS}[\widetilde{K}_L](1,2)=-\mathcal{I}_{SS}[\widetilde{K}_L](2,1)$ (with $\mathcal{I}_{SS}[\widetilde{K}_L](\alpha,\epsilon;\alpha,\epsilon)=0$).

3. *Because* $\cot(\phi_r/2)$ *has the same signs* $(+,-,+)$ *as* $\tan(\phi_r/2)$ *for all* $\alpha \in (0,\pi/3)$, $\sum_{r=1}^{3} c_r \operatorname{sgn}(w_r(\alpha,\epsilon)) = -2$ *for* ***both*** $\epsilon = \pm 1$. *Across all four orientations* $(\epsilon_1,\epsilon_2) \in \{+1,-1\}^2$ *and all three defect regimes* ($s<-1$, $|s|<1$, $s>1$, *with Levinson winding* $\Delta\theta_D(s,\eta) \in \{+2\pi\eta, 0, -2\pi\eta\}$*):*

$$I_{SS}^{\mathrm{raw}}(\alpha_1,\epsilon_1;\alpha_2,\epsilon_2) = \mathcal{I}_{SS}[\widetilde{K}_L](1,2) - \epsilon_1\frac{\pi}{2} = \mathcal{I}_{SS}[K_L](1,2) + \frac{1-\epsilon_1}{2}\pi, \tag{19}$$

$$J_{12}(\alpha_1,\epsilon_1;\alpha_2,\epsilon_2) := \frac{I_{SS}^{\mathrm{raw}}(1,2) - I_{SS}^{\mathrm{raw}}(2,1)}{2} = \mathcal{I}_{SS}[\widetilde{K}_L](1,2) - \frac{\epsilon_1-\epsilon_2}{4}\pi, \tag{20}$$

$$I_{SD}^{\mathrm{phys}}(\alpha,\epsilon;s,\eta) = \mathcal{I}_{SD}[\widetilde{K}_L](\alpha,\epsilon;q_D,\eta) + \frac{\epsilon}{4}\Delta\theta_D(s,\eta) = \mathcal{I}_{SD}[K_L](\alpha,\epsilon;q_D,\eta) - \frac{1-\epsilon}{4}\Delta\theta_D(s,\eta). \tag{21}$$

Equations (19)–(21) *explain the* $-\pi$ *two-soliton difference and* $+\eta\pi$ *stable-defect difference for* $\epsilon_1 = -1$ *found in* ***independent_phase2h_audit.pdf*** *Check 1, as well as* $I_{12}^{\mathrm{raw}} + I_{21}^{\mathrm{raw}} = -\frac{\pi}{2}(\epsilon_1+\epsilon_2)$ *in* ***audit_determinant_foundations.pdf****. Moreover,* $\mathcal{I}_{SS}[\widetilde{K}_L]$ *satisfies exact KP defect fusion and* ***zero*** *KP-triangle defect without any* $\pi/2$ *offset.*

2.2 Crossing Symmetry, Darboux Delay Fusion, and Breather Continuation

Proposition 2.2 (Crossing, Darboux coboundary, and breather continuation, **(c)**, `toda_wh_and_kp_synthesis`)

Crossing symmetry: *Under* $\alpha_2 \mapsto \alpha_2 + \pi$ *(which reverses the KP chord* $(q_{2,a}, q_{2,b}) \mapsto (q_{2,b}^{\mathrm{ho}}, q_{2,a}^{\mathrm{ho}})$ *and inverts* $A_{12}^{\mathrm{ov}} \mapsto 1/A_{12}^{\mathrm{ho}}$*), both the KP master functional and the canonical/mode renormalisation shifts* $\delta_{\mathrm{renorm}}(1,2) = \sum_{j=1}^{2} \Delta X_j'(1,2)\Delta P(j)$ *flip sign:*

$$\mathcal{I}_{SS}[\widetilde{K}_L]^{\mathrm{ov}}(\alpha_1,\alpha_2+\pi) = -\mathcal{I}_{SS}[\widetilde{K}_L]^{\mathrm{ho}}(\alpha_1,\alpha_2), \tag{22}$$

$$\delta_{\mathrm{renorm}}^{\mathrm{can/mode,ov}}(\alpha_1,\alpha_2+\pi) = -\delta_{\mathrm{renorm}}^{\mathrm{can/mode,ho}}(\alpha_1,\alpha_2), \tag{23}$$

while under spatial reflection $\alpha_2 \mapsto \pi - \alpha_2$ *at positive width* $k_2 > 0$*, both are strictly invariant.*

2. ***Darboux delay coboundary:*** *Defining the Darboux-shifted defect delay* $\Delta X_D'(1;q_D,\eta) := \Delta x_D(1;q_D,\eta) - 2\sqrt{3}\,\epsilon_1\eta\gamma_1 q_D$ *(*$\gamma_1 = d\Pi_1/dP_1 = -\frac{3\cos 2\alpha_1}{\sin\alpha_1 \sin 3\alpha_1}$*), the Darboux-shifted two-soliton delay is an exact 1-coboundary on the KP vertices:*

$$\Delta X_1'(1,2) = -\frac{\ln|A_{12}|}{k_1} + \gamma_1 M_2 = \epsilon_1\big[\Delta X_D'(1;q_{2,b},+1) - \Delta X_D'(1;q_{2,a},+1)\big]. \tag{24}$$

This pairs cleanly with the field-theoretic finding of ***audit_determinant_foundations.pdf*** *Eq. (8) that the asymptotic potential tail* $\phi_\alpha(+\infty) = 2\cos\alpha = M'(\alpha)$ *induces the cross-symplectic pairing* $\Omega(\partial_{\alpha_1},\partial_{\alpha_2}) \to -M_1'\Pi_2'$*, which is precisely the term removed by passing to Darboux coordinates* X_j'*.*

3. ***Analytical continuation to breather kinematics:*** *On the complex-conjugate breather locus* $\alpha_{1,2} = \bar\alpha \mp i\gamma$*, the* $z=+1$ *logarithmic branch points of* $K_{\chi_2}(z)$ *and* $K_{\ln}(z)$ *cancel identically in* $K_L(z)$ *(*$L'(z) = \frac{2\ln z}{1-z^2} \to -1$*), and*

$$\operatorname{Re}\mathcal{I}_{SS}[\widetilde{K}_L](\bar\alpha - i\gamma, \bar\alpha + i\gamma) = 0, \qquad \operatorname{Im}\big[\Delta P_{\mathrm{can}}(\bar\alpha - i\gamma) + \Delta P_{\mathrm{can}}(\bar\alpha + i\gamma)\big] = 0 \tag{25}$$

identically. Thus $\exp(i\mathcal{I}_{SS}[\widetilde{K}_L])$ *is a positive, pole-free real residue multiplier, while re-expressing* η *in physical rapidities shifts the Bogdan–Charkina quantisation condition* **(a)** *[3] to* $I_{\mathrm{br}}(\alpha_1^{\mathrm{phys}},\alpha_2^{\mathrm{phys}}) = \big[N - 2\operatorname{Im}(\frac{\partial I_{\mathrm{br}}}{\partial P_1}\Delta P(\alpha_1))\big]\hbar + O(\hbar^2)$*.*

2.3 Additional Audit Corrections and Open Field-Theoretic Questions

We record and verify in `toda_wh_and_kp_synthesis.py` the remaining algebraic corrections and physical qualifications established by the six audit reports:

1. **Separated head-on kernel asymptotics and soliton profile (`independent_phase2h_audit.pdf` Section 5) (c):** For $(\alpha_1, +1; \alpha_2, -1) = (r\lambda, +1; \lambda, -1)$ as $\lambda \to 0^+$, $I^{\text{head}}_{\chi_2} = \frac{9r}{\pi}\lambda^2 + O(\lambda^4)$ is purely analytic while all of the $\lambda^2 \ln(r\lambda^2)$ non-analyticity resides in $I^{\text{head}}_{\ln} = \frac{r\lambda^2}{\pi}\big[-3\ln(r\lambda^2) + 6\ln 2 - \frac{3}{2}\ln 3 - 12\big] + O(\lambda^4|\ln\lambda|)$. Under $\tau = 1 + e^{k(x-vt)}$ and $u = 2(\ln\tau)_{xx}$, the soliton profile is $u = +\frac{k^2}{2}\operatorname{sech}^2\frac{k(x-vt)}{2}$.

2. **Off-shell 4×4 defect sewing determinant and branch switch at $s = \sqrt{3}$ (`audit_frames_defects.pdf` Section 2) (c):** Away from its zeros, the 4×4 linearised Corrigan–Zambon sewing matrix **(a)** [2] satisfies

$$\frac{\det M}{(\kappa_1 - \kappa_2)^2} = -\frac{2}{3}S^2(S^2-1)P_-(S;s)P_+(S;s)$$

   (a factor of 16 larger than the $-1/24$ prefactor displayed in `oneloop_defect_phase.tex`). Because $P_-(1;s) = \frac{1}{2}(s+1)(s-\sqrt{3})$ and $P_+(1;s) = \frac{1}{2}(s-1)(s-\sqrt{3})$, at $s = \sqrt{3}$ the root $S_{-,+}$ drops below $S = 1$ while $S_{+,-}$ rises above $S = 1$. Thus across all $1 < s < 2$, the two admissible positive growth rates ($S > 1$) are $g_1(s) = \frac{s\sqrt{4-s^2}+\sqrt{3}(s^2-2)}{8} > 0$ and $g_2(s) = \frac{|s\sqrt{4-s^2}-\sqrt{3}(s^2-2)|}{8} \geq 0$, which coalesce at $g_1(2) = g_2(2) = +\frac{\sqrt{3}}{4}$.

3. **Scope of field-theoretic and defect identifications (d):** As documented in `audit_charges_action.pdf`, `audit_determinant_foundations.pdf`, `audit_frames_defects.pdf`, `audit_toda_dictionary.pdf`, and `audit_projected_measure.pdf`, the full fourth-order projected fluctuation determinant (including the non-compact translational/boost collective-coordinate measure and zero-mode subtraction, as well as the overall constant Z left open by Faddeev–Korepin **(a)** [1]), the boundary counterterm derivation for the fifth-degree Corrigan–Zambon defect potential, and the physical interpretation of KP-triangle fusion across unstable intermediate defect vacua remain open field-theoretic questions **(d)**.

3 Summary of Claims, Verification, and Failures Recorded

Claim	Label	Script / Check	Precision
Wiener–Hopf factorisation (3)–(5) and exact $\delta u(y)$ expansion (7) with $d_1 = e_2 = e_1 = 0$	**(c)**	`toda_wh_and_kp_synthesis.py` (`verify_1`)	Exact SymPy + $< 6\times 10^{-5}$
Krein convolution (13), exact $\ln Y$ cancellation, and proof of $R_0(\infty, r) = -\frac{4}{\pi}(\chi_2(r) - r)$	**(c)**	`toda_wh_and_kp_synthesis.py` (`verify_2`)	Exact SymPy + 70 digits
Closed-form two-loop coefficient $c_1(r)$ (16) and $\Delta c_1(r)$ (17)	**(c)**	`toda_wh_and_kp_synthesis.py` (`verify_2`)	Exact SymPy + $< 10^{-55}$
Classical-dispersion subtraction of saved Toda data matching exact $c_1(r)$	**(c)**	`toda_wh_and_kp_synthesis.py` (`verify_3`)	$< 2\times 10^{-4}$

Claim	Label	Script / Check	Precision
Inversion-antisymmetric kernel $\widetilde{K}_L(z)$ and exact 4-orientation identities (19)–(21)	**(c)**	`toda_wh_and_kp_synthesis.py` (`verify_4a`)	$< 10^{-64}$ (70 dps)
Crossing symmetry and Darboux delay coboundary $\Delta X_1'(1,2)$	**(c)**	`toda_wh_and_kp_synthesis.py` (`verify_4b`)	$< 10^{-64}$ (70 dps)
Breather continuation $\mathrm{Re}\,\mathcal{I}_{SS}[\widetilde{K}_L] = 0$ and $\mathrm{Im}\,\Delta P_{\mathrm{can,br}} = 0$	**(c)**	`toda_wh_and_kp_synthesis.py` (`verify_4c`)	$< 10^{-65}$ (70 dps)
Off-shell defect determinant ($-2/3$ prefactor) and $s = \sqrt{3}$ branch switch	**(c)**	`toda_wh_and_kp_synthesis.py` (`verify_4d`)	$< 10^{-60}$ (70 dps)
Fermi-edge reciprocity bilinear $8rY(\frac{1}{u'(Y)u'(r^2Y)} - 1) = \frac{4}{\pi}(r + \frac{1}{r}) + O(Y^{-1})$	**(c)**	`toda_wh_and_kp_synthesis.py` (`verify_4e`)	Exact SymPy + 70 digits
Full fourth-order projected determinant, zero-mode measure, defect UV counterterm, and all-loop S-matrix outside the Toda corner	**(d)**	Open field-theoretic questions	—

Failures recorded.

1. All prior-claim failures identified by the six independent audit reports (`independent_phase2h_audit.pdf`, `audit_charges_action.pdf`, `audit_determinant_foundations.pdf`, `audit_frames_defects.pdf`, `audit_toda_dictionary.pdf`, and `audit_projected_measure.pdf`) are reproduced and mathematically resolved above without modifying the audited files (whose SHA-256 hashes are verified by the audit scripts).

2. During the derivation of (16), an initial 2-integration-by-parts evaluation of the Krein convolution (13) expanded $\widetilde{\Psi}_-(v)$ asymptotically down to $v = 0$, where the large-v expansion is singular, producing a spurious offset in $c_1(r)$. Changing variables to the exact dressed momentum $v = u(\eta)$ ($\widetilde{\Psi}_-(v)\,dv = 2\sqrt{\eta}\,d\eta$ identically down to $\eta = 0$) removed the endpoint singularity and yielded exact agreement between the closed form (16) and 70-digit numerical quadrature to $< 10^{-55}$.

3. During testing of off-real-axis inversion $K(1/z) = -K(z)$ at complex $z = 0.85 + 0.23i$, $K_{\ln}(z) + K_{\ln}(1/z)$ alone gave $0.13213 \neq 0$ because $\ln\frac{1-1/z}{1+1/z} = \ln\frac{1-z}{1+z} \pm i\pi$ acquires a branch shift off the real axis that cancels against the $-\frac{i\pi}{2}\ln z$ term of $\chi_2(z) + \chi_2(1/z)$ inside the full kernel $\widetilde{K}_L(z)$.

References

[1] L. D. Faddeev and V. E. Korepin, *Quantum theory of solitons*, Phys. Rept. **42** (1978) 1–87, doi:10.1016/0370-1573(78)90058-3. (Full text in `literature/`.)

[2] E. Corrigan and C. Zambon, *Adding integrable defects to the Boussinesq equation*, J. Phys. A **56** (2023) 345203, arXiv:2306.03772. (Full text in `literature/`.)

[3] M. M. Bogdan and O. V. Charkina, *Quasiclassical quantization of the breather of the good Boussinesq equation*, Low Temp. Phys. **51** (2025), arXiv:2502.14695. (Full text in `literature/`.)

[4] M. Pustilnik and K. A. Matveev, *Phys. Rev. B* **92** (2015) 195146, arXiv:1507.05639. (Full text read in Phase 2d.)

[5] B. Sutherland, *Rocky Mountain J. Math.* **8** (1978) 413. (Cited in `toda_route.tex`.)

[6] F. G. Mertens, *Z. Phys. B* **55** (1984) 353. (Cited in `toda_route.tex`.)